🌞 Solar Panel Monitoring System

User Guide & Documentation

Version 1.0 | January 2026

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1. Introduction

What is the Solar Panel Monitoring System?

The **Solar Panel Monitoring System** is a professional-grade web application designed to help solar panel owners, installers, and engineers analyze the performance of photovoltaic (PV) systems. Whether you have a small residential setup or a commercial installation, this tool provides comprehensive insights into your solar energy production.

Key Benefits

Benefit	Description
Performance Tracking	Monitor daily and monthly energy production
Efficiency Analysis	Calculate Performance Ratio (PR) to understand system efficiency
Problem Detection	Identify underperformance through Expected vs. Actual comparisons
Temperature Impact	Understand how heat affects your panel efficiency
Battery	Filter data when batteries are full for accurate PR

Awareness	calculations
Professional Reports	Export high-resolution graphs for documentation

2. Getting Started

Accessing the Application

Open your web browser and navigate to the application URL:

```
http://localhost:8501
```

You will see the main dashboard interface:

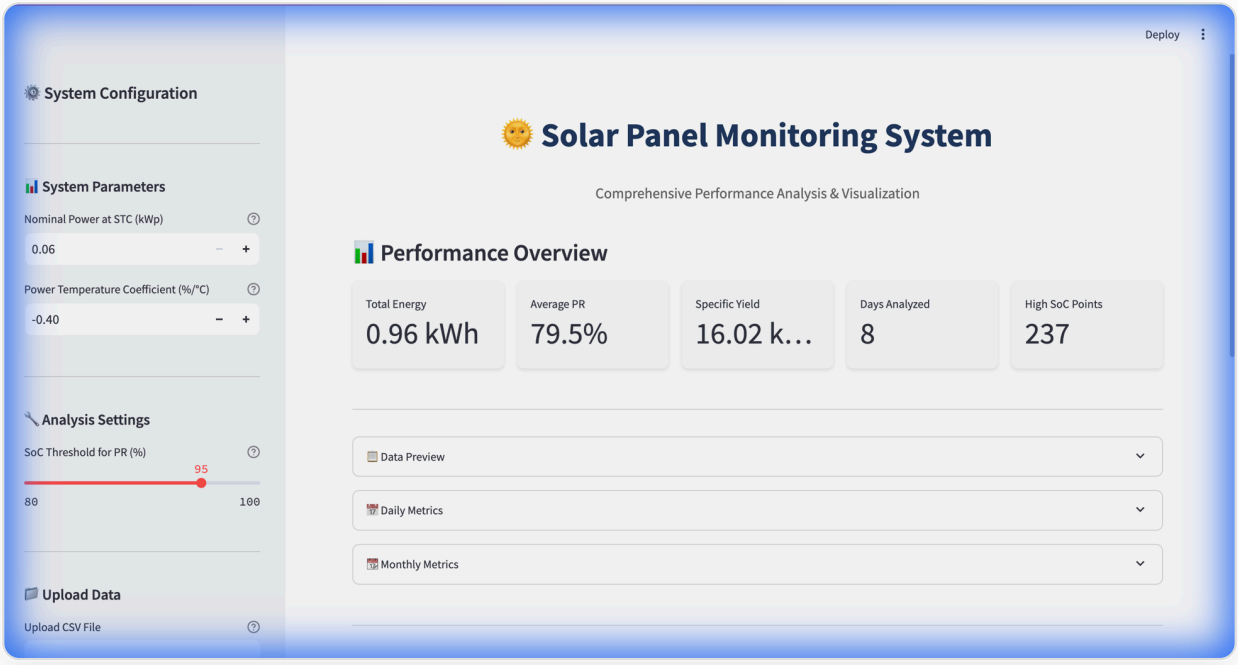


Figure 1: Main Dashboard - Performance Overview with key metrics

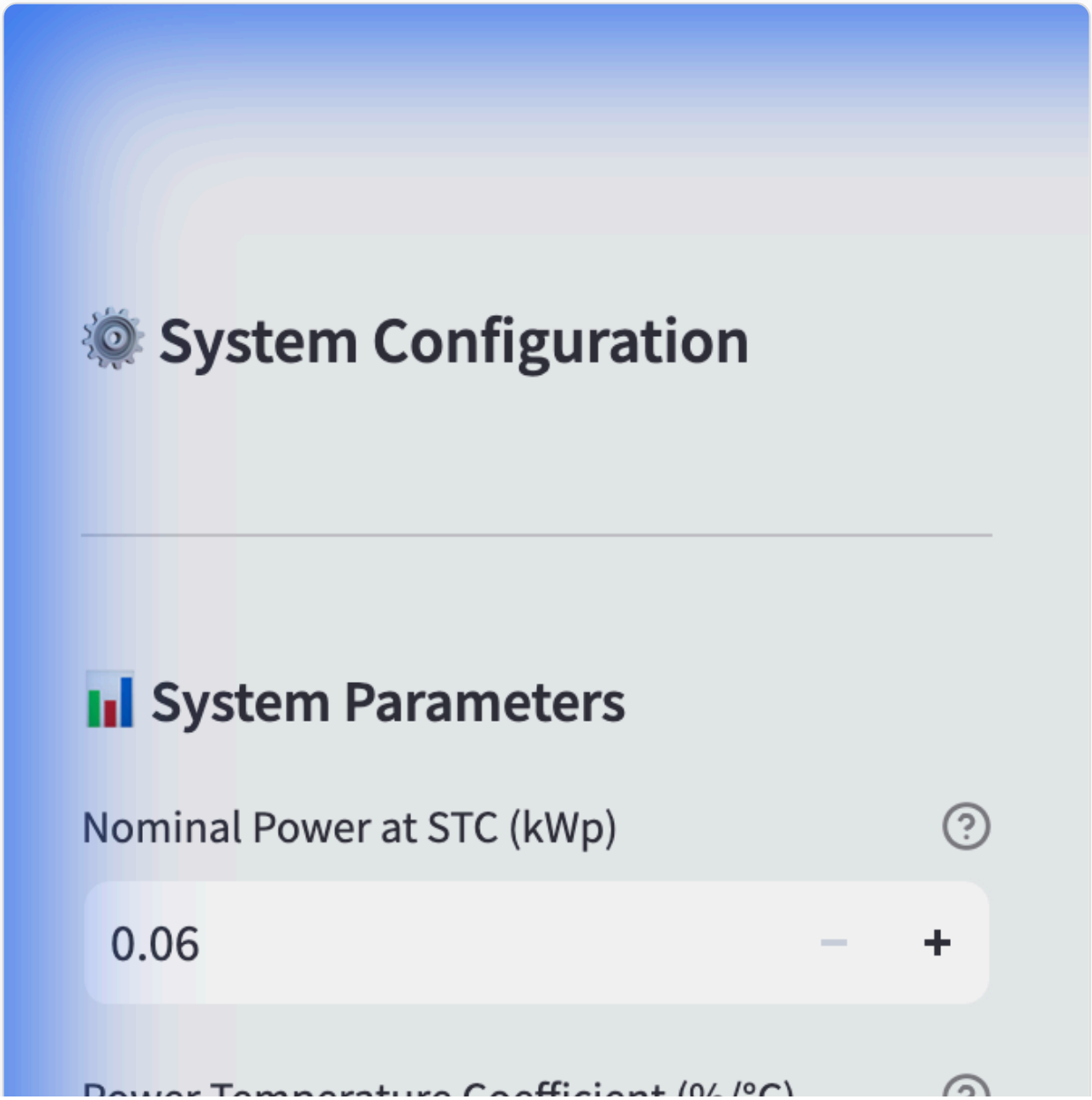
The interface consists of two main areas:

- **Left Sidebar** – System configuration and data upload
- **Main Content Area** – Performance metrics, graphs, and analysis

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3. System Configuration

Before analyzing your data, you need to configure your solar system parameters. These settings are located in the **left sidebar**.



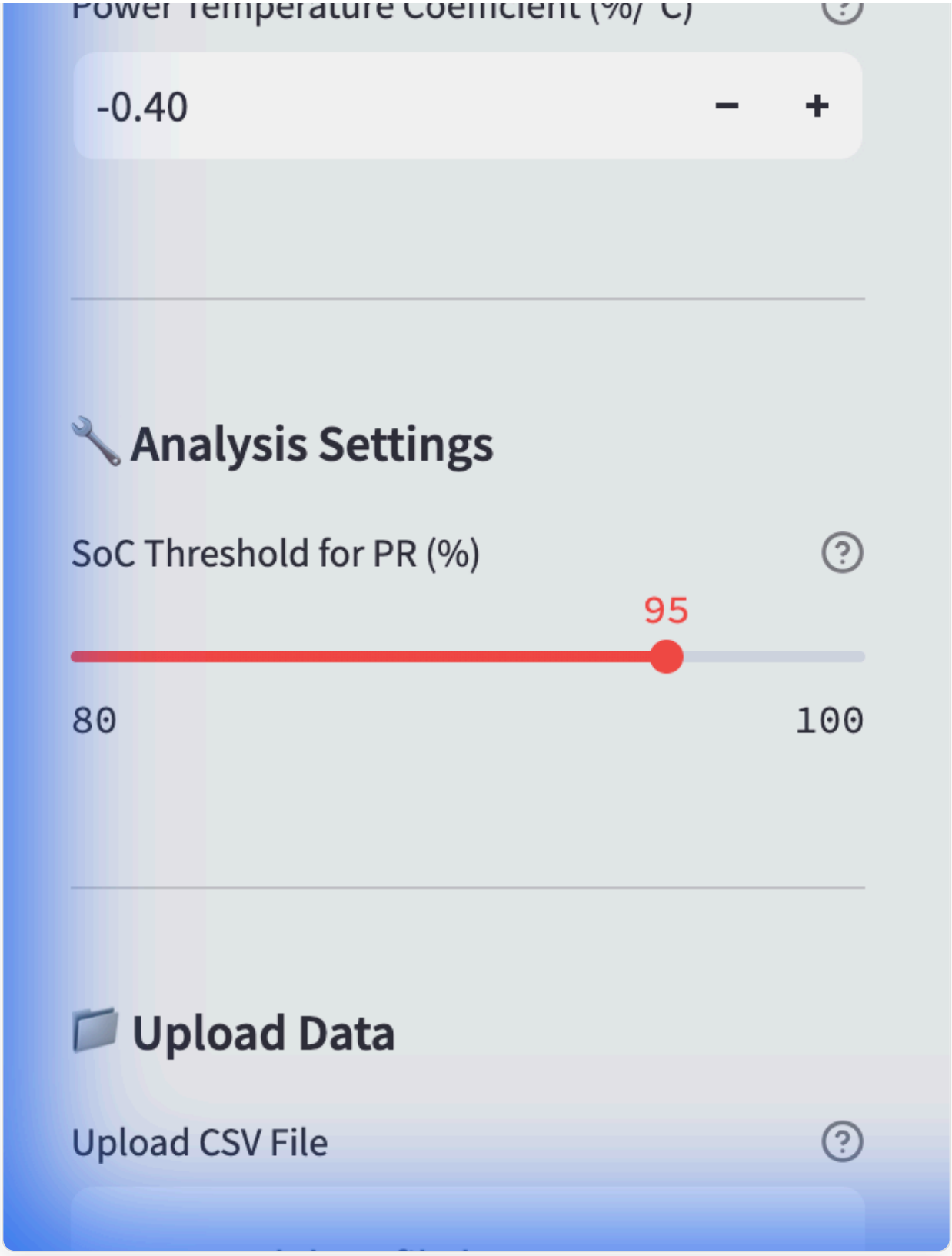


Figure 2: System Configuration Panel - Enter your system parameters here

3.1 System Parameters

Nominal Power at STC (kWp)

This is your solar system's **rated power output** under Standard Test Conditions.

Field	Description
What to Enter	Your panel/system's nameplate power rating
Unit	Kilowatt-peak (kWp)
Allowed Range	0.001 to 1000 kWp
Example	For a 48W panel, enter `0.048`

> 💡 **Tip: How to Find This Value**
>
> Check your solar panel's datasheet or label. Look for "Pmax" or "Nominal Power" at STC (1000 W/m², 25°C).

Power Temperature Coefficient (%/°C)

This value represents how much your panel's power output decreases as temperature rises above 25°C.

Field	Description
What to Enter	Temperature coefficient from your panel's datasheet
Unit	Percentage per degree Celsius
Typical Range	-0.30% to -0.50% for silicon panels
Example	Enter `-0.40` for a -0.40%/°C coefficient

3.2 Analysis Settings

SoC Threshold for PR (%)

This setting controls how the system handles data when your **battery is nearly full**.

> ⚠️ **Important: Why This Matters**
>
> When your battery reaches high State of Charge (SoC), the solar system often
> **curtails (reduces) power generation**
> to protect the battery. Including this curtailed data in PR calculations would incorrectly suggest poor performance. The system automatically excludes these periods from PR calculations while still including them in total energy counts.

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4. Uploading Your Data

The system accepts solar monitoring data in **CSV format**. You have two options for providing data:

4.1 Upload a CSV File

1. In the sidebar, locate the **"Upload CSV File"** section
2. Click **"Browse files"** or drag and drop your CSV file
3. Wait for the file to be processed

4.2 Paste Data Directly

1. Scroll down in the sidebar to **"Or paste data directly"**
2. Copy your CSV data (including headers)

3. Paste into the text area

4.3 Process Your Data

1. Click the "🚀 **Process Data**" button
2. Wait for the system to analyze your data
3. View the **Column Detection Results** to confirm all columns were identified

Status	Meaning
✅ Detected 'Column Name' as type	Column successfully identified
⚠️ Could not detect type column	Column not found (optional columns may be missing)
🔄 Detected mV, converting to V	Automatic unit conversion applied
📊 Data frequency: ~X minutes	Data interval detected

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5. Understanding the Dashboard

Once your data is processed, the dashboard displays key performance metrics.

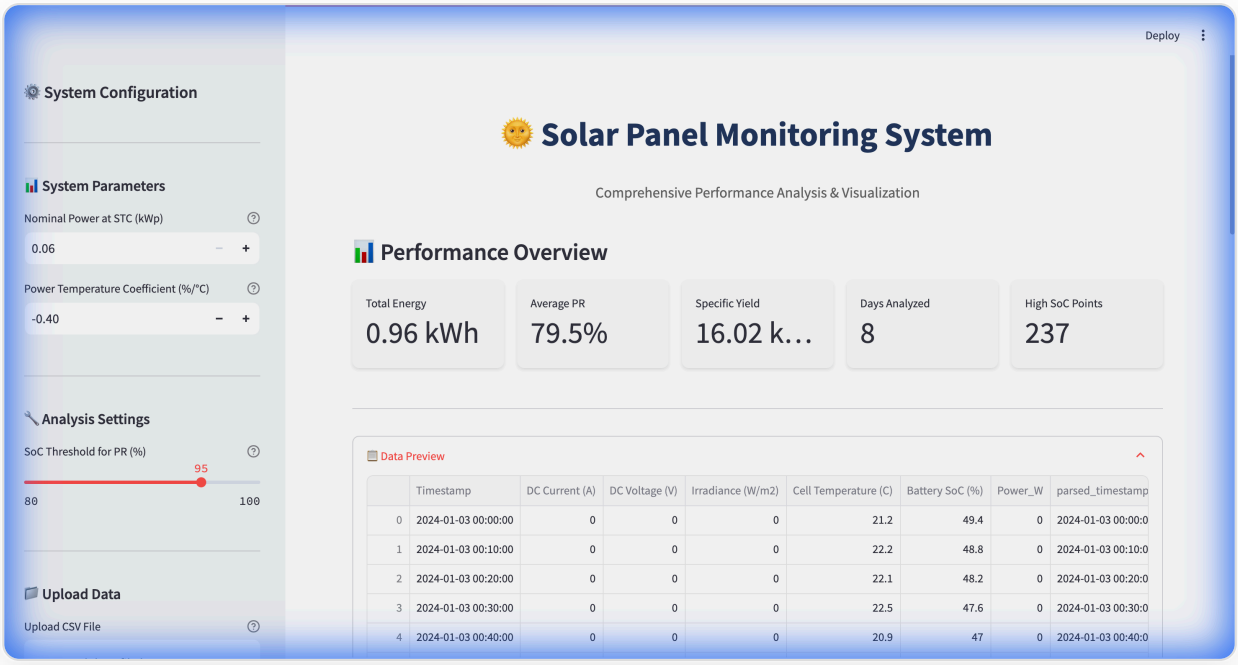


Figure 3: Performance Overview with Data Preview showing calculated power values

5.1 Performance Overview Cards

The top section displays five key performance indicators:

Metric	Description	Good Values
Total Energy	Total electricity generated (kWh)	Higher is better
Average PR	Average Performance Ratio across all days	75-85% typical
Specific Yield	Energy per installed capacity (kWh/kWp)	3-6 kWh/kWp/day
Days Analyzed	Number of days in your dataset	Based on your data
High SoC Points	Data points where battery was ≥95% full	Excluded from PR

5.2 Data Preview Section

Expand the **"Data Preview"** section to see your raw data with calculated fields including Power_W (Current × Voltage).

6. Daily Performance Analysis

Navigate to the 📈 **Daily Graphs** tab to view detailed daily performance charts.

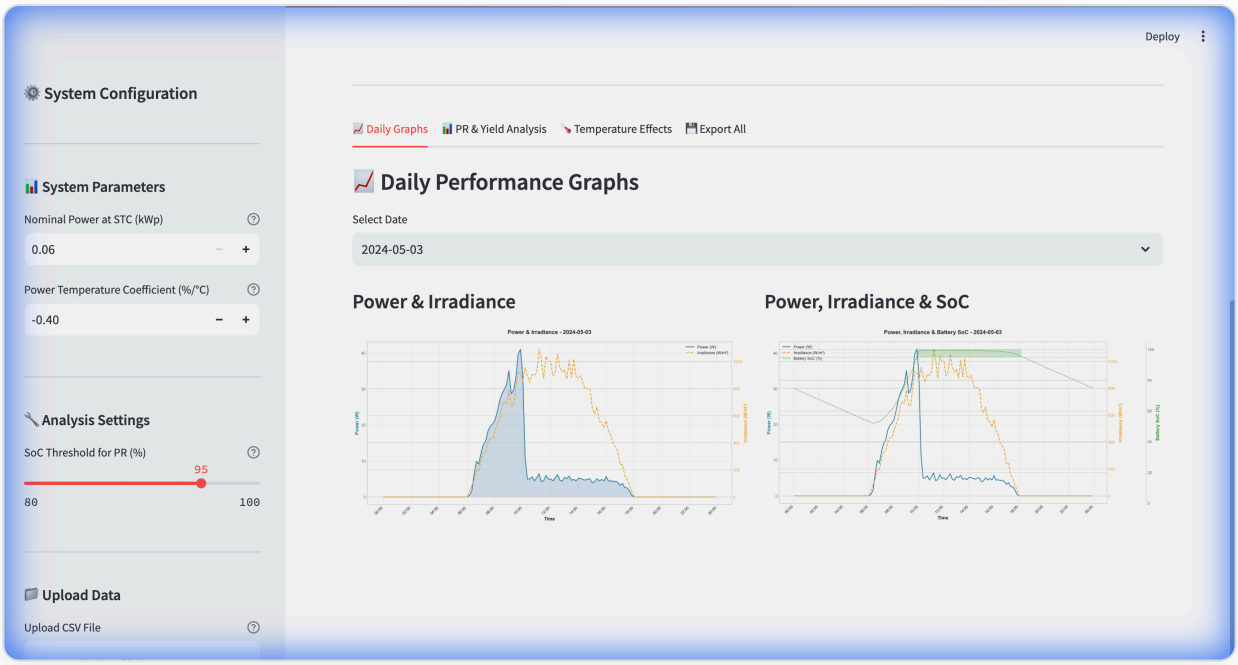


Figure 4: Daily Performance Graphs - Power & Irradiance with Battery SoC tracking

6.1 Selecting a Date

Use the **"Select Date"** dropdown to choose which day's data to visualize. Each day has its own separate graphs.

6.2 Power & Irradiance Graph (Left)

This chart shows the relationship between solar radiation and your system's power output throughout the day.

Line Color	Metric	Description
Blue (filled area)	Power (W)	Your system's actual power output
Orange (dashed)	Irradiance (W/m²)	Solar radiation hitting the panels

6.3 Power, Irradiance & SoC Graph (Right)

This three-way chart adds battery state-of-charge to the analysis, showing when power curtailment may occur.

> 📝 **Note**
>
> When SoC reaches 95%+ (shown in green), the system may be curtailing power output. This data is excluded from Performance Ratio calculations.

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7. Performance Ratio & Yield Analysis

Navigate to the 📊 **PR & Yield Analysis** tab for long-term performance trends.

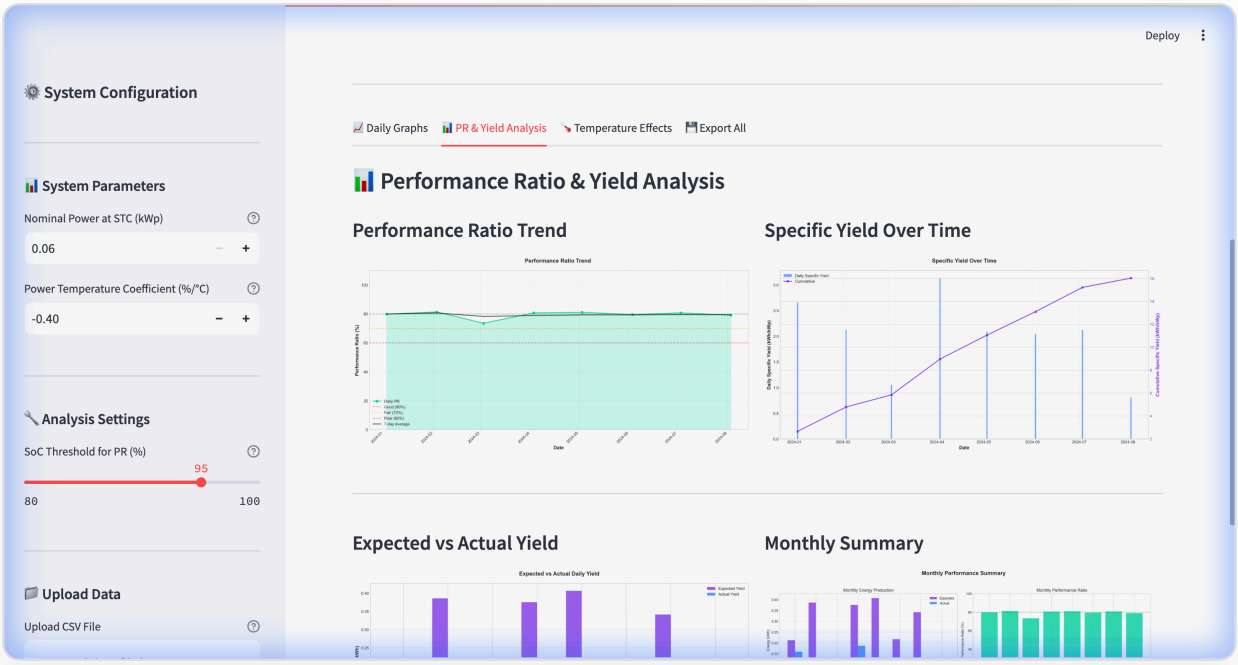


Figure 5: PR & Yield Analysis - Trend charts, Expected vs Actual, and Monthly Summary

7.1 Performance Ratio Trend (Top Left)

This chart shows how your system's efficiency changes over time.

PR Range	Performance Level	Recommended Action
80%+	Excellent	System performing optimally
70-80%	Good	Normal operation
60-70%	Fair	Investigate potential issues
Below 60%	Poor	Immediate inspection recommended

7.2 Specific Yield Over Time (Top Right)

Shows energy production per unit of installed capacity with cumulative tracking.

7.3 Expected vs. Actual Yield (Bottom Left)

Compares theoretical maximum output against actual production to identify underperformance.

7.4 Monthly Summary (Bottom Right)

Four-panel summary showing monthly trends for energy, PR, yield, and conditions.

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8. Temperature Effect Analysis

Navigate to the 🌡️ **Temperature Effects** tab to understand how heat impacts your panels.

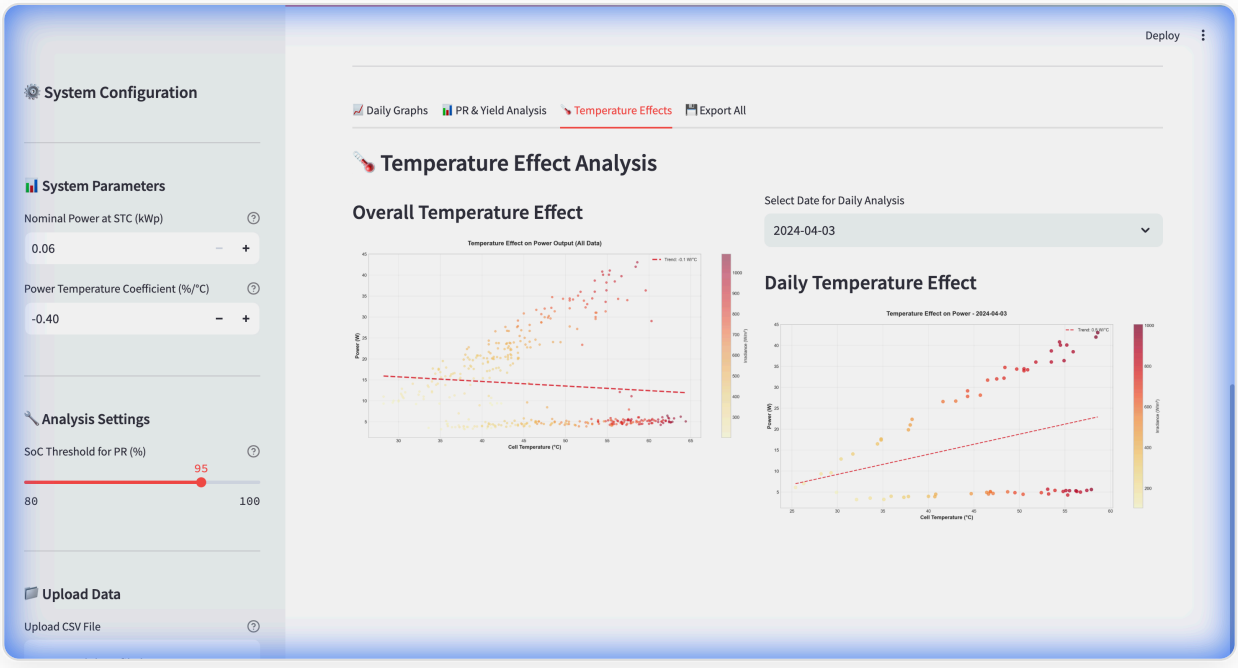


Figure 6: Temperature Effect Analysis - Scatter plots showing power vs. temperature correlation

8.1 Overall Temperature Effect (Left)

This scatter plot shows power output vs. cell temperature for your entire dataset.

- Each **dot** represents one data reading
- **Color gradient** indicates irradiance (yellow/red = higher)
- **Trend line** shows the temperature effect coefficient

8.2 Daily Temperature Effect (Right)

Select a specific date to analyze temperature effects for that day only.

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9. Exporting Data & Reports

Navigate to the 💾 **Export All** tab to generate and download reports.

9.1 Generating All Graphs

Click "📥 **Generate All Graphs**" to create high-resolution images:

Graph Type	Filename Format
Power & Irradiance (Daily)	`YYYY-MM-DD_Power_Irradiance.png`
Power, Irradiance & SoC (Daily)	`YYYY-MM-DD_Power_Irradiance_SoC.png`
Temperature Effect (Daily)	`YYYY-MM-DD_Temperature_Effect.png`
PR Trend	`DateRange_PR_Trend.png`
Specific Yield	`DateRange_Specific_Yield.png`

Expected vs Actual	`DateRange_Expected_vs_Actual.png`
Monthly Summary	`Monthly_Summary.png`

9.2 Image Specifications

Property	Value
Resolution	1400 × 700 pixels
DPI	150 (high resolution)
Format	PNG (lossless)

9.3 Download Options

- 📦 **Download All as ZIP** – All generated graphs in one archive
- 📥 **Download Processed Data (CSV)** – Full dataset with calculated fields
- 📥 **Download Daily Metrics (CSV)** – Day-by-day performance summary
- 📥 **Download Monthly Metrics (CSV)** – Month-by-month summary

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10. Understanding Key Metrics

10.1 Performance Ratio (PR)

Definition: The ratio of actual energy output to the theoretical maximum output.

```
PR = (Actual Energy / Expected Energy) × 100%
```

> ⚠️ **High SoC Exclusion**
>
> The system automatically excludes periods when Battery SoC ≥ 95% from PR calculations. This prevents curtailed generation from artificially lowering your PR score.

10.2 Specific Yield

Definition: Energy generated per unit of installed capacity.

```
Specific Yield = Energy (kWh) / Installed Capacity (kWp)
```

10.3 Expected Power

Definition: Theoretical power output based on current conditions.

```
Expected Power = Rated Power × (Irradiance / 1000) × Temperature

Temperature Factor = 1 + (Temp Coefficient × (Cell Temp - 25°C))
```

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11. Data Format Requirements

11.1 Required and Optional Columns

Column	Status	Description	Example

Timestamp	Required	Date and time	`2024-03-01 10:30`
Current	Required	DC current (A)	`5.2`
Voltage	Required	DC voltage (V)	`32.5`
Irradiance	Required	Solar radiation (W/m²)	`850`
Temperature	Optional	Cell temp (°C)	`35`
Battery SoC	Optional	Charge level (%)	`85`

11.2 Sample Data Format

```
Timestamp,DC Current (A),DC Voltage (V),Irradiance (W/m2),Cell Te
2024-03-01 06:00,0.5,28.5,50,22,45
2024-03-01 06:30,1.2,29.1,120,23,46
2024-03-01 07:00,2.5,30.2,250,24,48
```

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12. Troubleshooting

"Please upload a file or paste data to process"

Solution: Upload a CSV file or paste data, then click "🚀 Process Data"

Power values showing "None" or 0

Solution: Check that your column headers match expected formats. Verify column detection shows ✅ for current and voltage.

Performance Ratio showing very low values

Solution: Verify your Nominal Power setting is in kWp (not W). Check if voltage was auto-converted from mV.

No graphs appearing

Solution: Ensure your data has all required columns (Timestamp, Current, Voltage, Irradiance) and at least one full day of readings.

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13. Glossary

Term	Definition
DC Current	Direct current flowing from solar panels (Amperes)
DC Voltage	Direct current voltage from solar panels (Volts)
GHI	Global Horizontal Irradiance – total solar radiation on horizontal surface
Irradiance	Solar power per unit area (W/m²)
kWh	Kilowatt-hour – unit of energy
kWp	Kilowatt-peak – rated power under STC
PR	Performance Ratio – system efficiency measure
SoC	State of Charge – battery level (%)

STC	Standard Test Conditions – 1000 W/m², 25°C
Specific Yield	Energy per capacity (kWh/kWp)

☀️ Solar Panel Monitoring System v1.0

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